Chongjing XIA
PhD in Plant Pathology

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PROFESSIONAL EXPERIENCES

30/01/2024-Present	Bioinformatician/Research Associate	University of Cambridge, Cambridge, UK
06/23/2020-12/31/2023	Associate Professor	Southwest University of Science and Technology, China
09/20/2018-06/22/2020	Lecturer	Southwest University of Science and Technology, China
08/25/2014-07/10/2018	PhD Student	Washington State University, USA
01/02/2012-07/17/2014	MSc Student	Washington State University, USA

EDUCATION

07/10/2018	PhD in Plant Pathology Washington State University, USA
07/17/2014	MSc in Plant Pathology Washington State University, USA
07/01/2011	BSc in Plant Protection Northwest A&F University, China

FUNDINGS

- 01/01/2020-12/31/2022, High-density genetic map construction and mapping of *Avr* genes in the wheat stripe rust fungus, issued by the National Natural Science Foundation of China, ¥250,000, **Independent Principal-Investigator**
- 01/01/2021-12/31/2022, Genomic structure variation detection and application in wheat stripe rust fungus, issued by State Key Laboratory for Biology of Plant Diseases and Insect Pests, ¥50,000, **Independent Principal-Investigator**
- 01/01/2019-12/31/2021, Population genomics of wheat stripe rust pathogen in SiChuan Province, issued by Southwest of University of Science and Technology, ¥250,000, **Independent Principal-Investigator**
- 01/12/2021-11/30/2024, Virulence variation and population adaptability of the wheat stripe rust fungus, sub-project of the National Key R&D Program, issued by Ministry of Science and Technology, ¥4,600,000, Co-Principal-Investigator

MAJOR RESEARCH RESULTS

- Developed an enhanced attention-based deep learning model to classify disease severity and infection type of wheat stripe rust (accuracy>95%);
- Revealed the structural features of 3D genomes of plant pathogenic fungi using multi-omics analysis;
- Independently assembled three sets of stripe rust fungal genomes to the chromosome-level, and one tetraploid plant genome to haplotype-resolved level;
- Unraveled the genomic basis of the evolution of wheat stripe rust pathogenic fungi;
- Developed a new algorithm for genome-wide association analysis and screened multiple virulence phenotype-associated genes;
- Established bioinformatics analysis pipelines: RNA-seq analysis, GWAS, single-cell transcriptomics, etc.

PUBLICATIONS (†Equal contribution, *Correspondence)

First & Corresponding author publications

Wu Y†, Shen L†, Zhang XZ, Huang L, Liu WZ, Chen WQ, Liu TG*, **Xia CJ***. 2026. An enhanced attention-based deep learning model enables accurate assessment of disease severity and infection type of wheat stripe rust. (in preparation)

Xu H, Wang S, Li X, Zhou X, Chen W*, **Xia CJ***. 2026. A Comprehensive Benchmarking of 16 Long-read Sequencing-Based Genome Assemblers in Filamentous Fungi. (BMC Biology, under review)

Li X, Tan W, Feng J, Yan Q, Tian R, Chen Q, Li Q, Zhong S, Yang S, **Xia CJ***; Zhou XL*. 2025. Mapping QTLs for Stripe Rust Resistance and Agronomic Traits in Chinese Winter Wheat Lantian 31 Using 15K SNP Array. **Agriculture** 15:1444.

Huang J†, Zhang X†, Tan W†, Wu Y, Wang S, Mehmood S, Zhou X, Yang S, Wang M, Chen X, Chen WQ, Liu TG, Li X*, **Xia CJ.*** 2025. Virulence Characterization of *Puccinia striiformis* f. sp. *tritici* in China in 2020 Using Wheat *Yr* Single-Gene Lines. **Journal of Fungi**. 11(6):447

Xing Y†, **Xia CJ**†, Huang L, Zhao N, Li H, Zhang X, Qiu AG, Tang WQ, Wang MN, Chen XM, Liu B, Zhang H, Gao L, Chen WQ*, Liu TG.* 2025. Population genomic analyses reveal extensive genomic regions within selective sweeps associated with adaptation and demographic history of a wheat fungal pathogen. **New Plant Protection**. https://doi.org/10.1002/npp2.70003

Zhang J, Zhao W, Tang X, Jia G, Li D*, **Xia CJ*,** Bai SQ.* 2025. Chromosome-level genome and annotation of the tetraploid *Rhodiola kirilowii*. **Scientific Data**. 12, 620

Zhang X, Huang L, Tang W, Qiu A, Li X, Zhou X, Yang S, Wang M, Chen XM, Chen WQ, Liu TG*, **Xia CJ*** 2024. Virulence Characterization of *Puccinia striiformis* f. sp. *tritici*, the Causal Agent of Wheat Stripe Rust in China in 2023 Using the 18 *Yr* Single-gene Differentials. **Plant Pathology**. 74:363-377.

Xia CJ†, Huang L†, Huang J, Zhang H, Huang Y, Benhamed, Wang MN, Chen XM, Zhang M, Chen WQ, and Liu TG. 2022. Folding features and dynamics of 3D genome architecture in plant fungal pathogens. **Microbiology Spectrum**. 10(6):e0260822.

Xia CJ*, Qiu AG, Wang MN, Liu TG, Chen WQ, Chen XM.* 2022. Current status and future perspectives of genomics research in the rust fungi. **International Journal of Molecular Sciences**. 23(17):9629.

Liu TL, Bai Q, Wang MN, Li YX, Wan AM, See DR, **Xia CJ.***, and Chen XM.* 2021. Genotyping *Puccinia striiformis* f. sp. *tritici* isolates with SSR and SP-SNP markers reveals dynamics of the wheat stripe rust pathogen in the United States from 1968 to 2009 and identifies avirulence associated markers. **Phytopathology.** 111:1828-1839.

Xia CJ†, Lei Y†, Wang MN, Chen WQ*, and Chen XM.* 2020. An avirulence gene cluster in the wheat stripe rust pathogen (*Puccinia striiformis* f. sp. *tritici*) identified through genetic mapping and whole-genome sequencing of a sexual population. **mSphere** 5:e00128-20.

Xia CJ, Wang M, Yin C, Cornejo OE, Hulbert SH, and Chen XM.* 2018. Genomic insights into host adaptation between the wheat stripe rust pathogen (*Puccinia striiformis* f. sp. *tritici*) and the barley stripe rust pathogen (*Puccinia striiformis* f. sp. *hordei*). **BMC Genomics**. 19:664.

Xia CJ, Wang M, Yin C, Cornejo OE, Hulbert SH, and Chen XM.* 2018. Genome sequence resources for the wheat stripe rust pathogen (*Puccinia striiformis* f. sp. *tritici*) and the barley stripe rust pathogen (*Puccinia striiformis* f. sp. *hordei*). **Molecular Plant-Microbe Interactions.** 31:1117-1120.

Xia CJ, Wang MN, Cornejo OE, Jiwan DA, See DR, and Chen XM.* 2017. Secretome characterization and correlation analysis reveal putative pathogenicity mechanisms in the wheat stripe rust fungus *Puccinia striiformis* f. sp. *tritici*. **Frontiers in Microbiology.** 8:2394.

Xia CJ, Wan AM, Wang MN, Jiwan DA, See DR, and Chen XM.* 2016. Secreted protein gene derived-single nucleotide polymorphisms (SP-SNPs) reveal population diversity and differentiation of *Puccinia striiformis* f. sp. *tritici* in the United States. **Fungal Biology**. 120:729-744.

Co-author publications

Hu Y, Li M, Li Y, Du L, Xie R, Ni F, **Xia CJ**, Wang K, Huang Y, Xu B, Li Y, Jiang Y, Hao M, Jiang B, Ning S, Yuan Z, Feng L, Zhang L, Chen S, Wu B*, Liu Z*, Fahima T*, Liu D*, Huang L*. A head-to-head NLR gene

pair from wild emmer confers stripe rust resistance in wheat. **Nature Genetics**. 2025 Jun 9. doi: 10.1038/s41588-025-02207-0.

Huang, L., Zhao, W., **Xia, C.J.**, Zhao, N., Li, H., Sun, Z., Yang, L., Li, M., Chen, W., Yang, F., Zhang, H., Chen, W., Liu, T-G. Wheat Stripe Rust Inoculum from the Southwest Dispersed to the East Huang-Huai-Hai Region Through Southern Anhui in China. Plant Disease 2025. Jan 13:PDIS06241246RE. doi: 10.1094/PDIS-06-24-1246-RE. Epub ahead of print. PMID: 39182161.

Yan, D., Huang, L., Xing, Y., Li, H., **Xia, C.J.**, Wang, A. Gao., L., Liu, T., Chen, W.Q. 2023. Analysis of genetic diversity and population structure of *Puccinia striiformis* f. sp. *tritici* infers inoculum relationships from Yunnan to the middle and lower reaches of the Yangtze River. Plant Pathology, 72, 1517-1527.

Huang, Y., An, J., Sircar, S., Bergis, C., Lopes, C.D., He, X., Da Costa, B., Tan, F.Q., Bazin, J., Antunez-Sanchez, J., Mammarella, M.F., Devani, R.S., Brik-Chaouche, R., Bendahmane, A., Frugier, F., **Xia, C.J.**, Rothan, C., Probst, A.V., Mohamed, Z., Bergounioux, C., Delarue, M., Zhang, Y., Zheng, S., Crespi, M., Fragkostefanakis, S., Mahfouz, M.M., Ariel, F., Gutierrez-Marcos, J., Raynaud, C., Latrasse, D., Benhamed, M. 2023. HSFA1a modulates plant heat stress responses and alters the 3D chromatin organization of enhancer-promoter interactions. **Nature Communications**. 14(1):469.

Li, Y., Hu, Y., Jiang, Y., Zhou, Q., He, Y., He, J., Chen, X., Chen, X., Jiang, B., Hao, M., Ning, S., Yuan, Z., Zhang, J., **Xia, C.J.**, Wu, B., Feng, L., Zhang, L., Liu, D., & Huang, L. 2023. Identification and fine-mapping of QYrAS286-2BL conferring adult-plant resistance to stripe rust in cultivated emmer wheat. Theoretical and Applied Genetics. Theoretical and Applied Genetics, 137(1):5.

Zhou, J., Zheng, X., Zhong, X., Tan, W., Ma, C., Wang, Y., Tian, R., Yang, S., Li, X., **Xia, C.J.**, Kang, Z., Chen, X., Zhou, X. 2023. Transfer of the high-temperature adult-plant stripe rust resistance gene *Yr62* in four Chinese wheat cultivars. Molecular Breeding. 22;43(6):44.

Zheng, X., Zhou, J., Zhang, M., Tan, W., Ma, C., Tian, R., Yan, Q., Li, X., **Xia, C.J.**, Kang, Z., Chen, X., Zhou, X., Yang, S. 2022. Transfer of Durable Stripe Rust Resistance Gene *Yr39* into Four Chinese Elite Wheat Cultivars Using Marker-Assisted Selection. Agronomy. 12:1791.

Bai, Q., Wang, M., **Xia, C.J.**, See, D.R., and Chen, X.M. 2022. Identification of secreted protein gene-based SNP markers associated with virulence phenotypes of *Puccinia striiformis* f. sp. *tritici*, the wheat stripe rust pathogen. International Journal of Molecular Science. 23(8):4114.

Huang, L., Yang, H., **Xia, C.J.**, Li, H., Wang, J., Wang, A., Zhang, M., Kang, X., Gao, L., Zhou, Y., Chen, W.Q., and Liu, T.G. 2022. Long-distance transport of *Puccinia striiformis* f. sp. *tritici* by upper airflow on the Yunnan-Guizhou Plateau disrupts the balance of agricultural ecology in central China. **Plant Disease**. 106(11):2940-2947.

Li, Y.X., **Xia, C.J.**, Wang, M.N., Yin, C., and Chen, X.M. 2020. Whole-genome sequencing of *Puccinia striiformis* f. sp. *tritici* mutant isolates identifies avirulence gene candidates. **BMC Genomics.** 21:247.

PRESENTATIONS

Oral presentations

2023 Plant Pathological Society of SiChuan Province annual meeting, Chengdu, China. Chromosome-level genome reference of CYR34 and avirulence gene identification.

2020 China Society of Plant Protection annual meeting, Guiyang, China. Avirulence gene identification in wheat stripe rust fungus via genetic mapping and whole-genome sequencing.

2019 Plant Pathological Society of SiChuan Province annual meeting, Chengdu, China. Genomic basis of host adaptation of cereal rust fungi.

2017 American Phytopathological Society Pacific Division Annual Meeting, Riverside, CA, USA. Secretome characterization and correlation analysis reveal putative pathogenicity mechanisms in the wheat stripe rust fungus *Puccinia striiformis* f. sp. *tritici*.

2015 American Phytopathological Society Annual Meeting, Pasadena, CA, USA. Secreted protein gene derived-single nucleotide polymorphisms (SP-SNPs) reveal population diversity and differentiation of *Puccinia striiformis* f. sp. *tritici* in the United States.

Poster presentations

2023 12th International Conference of Plant Pathology, Lyon, France. Folding features and dynamics of 3D genome architecture in plant fungal pathogens.

2018 11th International Conference of Plant Pathology, Boston, MT, USA. Genomic insights into host adaptation between the wheat stripe rust pathogen (*Puccinia striiformis* f. sp. *tritici*) and the barley stripe rust pathogen (*Puccinia striiformis* f. sp. *hordei*).

HONORS & AWARDS

2022 First Prize, Science and Technology Award, Chinese Society of Plant Protection
2019 High-Level Overseas Talent of Sichuan Province
2017 APS Pacific Division Student Travel Award
2015 Washington State University Graduate and Professional Student Travel Grant
2011 Chinese Government Scholarship for Overseas Study, China Scholarship Council

TEACHING

2019-2023 General Plant Pathology, (undergraduates)
2020-2023 Topics in Plant-Pathogen Interactions (MSc)
2020-2023 Topics in Advanced Genomics in Wheat Research (PhD)

MENTORSHIP

Hai Xu:

2022-2024, MSc student, Southwest of University of Science and Technology, solely under my supervision

Project: benchmarking third-generation genome assemblers on fungi

Xingzong Zhang:

2021-2023, MSc student, Southwest of University of Science and Technology, solely under my supervision

Project: virulence characterization of wheat stripe rust in China in 2023

Wanqiang Tang:

2020-2022, MSc student, Southwest of University of Science and Technology, solely under my supervision

Project: virulence characterization of wheat stripe rust in China in 2021

Liang Huang:

2018-2022, PhD student, Chinese Academy of Agricultural Sciences Institute of Plant Protection, co-supervision with Professor Wanquan Chen

Project: Long-distance transport of *Puccinia striiformis* f. sp. *tritici* in central China

PROFESSIONAL SERVICE

Society membership

2022-2024 Director of Sichuan Society of Plant Protection
2022-2024 Director of Sichuan Society of Plant Pathology
2021-2024 Executive Member, Smart Agriculture Branch, Sichuan Society for Bioinformatics
2019-2024 Member of China Society of Plant Protection
2015-2024 Member of American Phytopathological Society

Editorial & peer reviewer

2022 Guest Editor, Frontiers in Genetics

2018-2024 Peer reviewer for New Phytologist, Molecular Plant Pathology, Plant Disease, Scientific Data, Food and Energy Security, Journal of Integrative Agriculture, Microbiology Spectrum, Frontiers in Microbiology, BMC Microbiology, Frontiers in Plant Science, Environmental Microbiology, NAR Genomics and Bioinformatics, Frontiers in Genetics, Plant Pathology, Journal of Plant Pathology, PLOS ONE, Crop Protection

Resources

https://github.com/chongjing/Linux101

https://github.com/chongjing/scRNAseq_Medicago